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Study of 100 kHz femtosecond laser continuous scanning welding and spot welding of glass and aluminium film

[Hongrui Wang](#), [Wenchao Zhong](#), [Zizheng Zhao](#), [Bowen Liu](#), [Dongqing Pang](#)  , [Minglie Hu](#) [Show more](#) ▼[View PDF](#)[Download full issue](#)[Cite](#)[Add to Mendeley](#)[Share](#)[10.1016/j.optlastec.2025.113969](#) ↗[Article](#)[Recommended articles](#)[Metrics](#)

Outline

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Highlights

- High density welding region and high weld strength can be obtained by 100 kHz fs laser welding.
- Single pulse dynamics is dominant, so that controlling energy density at the interface is most important.
- Filamentation acts like nonlinear lens and welding dynamics needs to include effect of shock wave.
- Mixture of plasma is principal reason of strong weld strength and gradually varying material properties may be beneficial.

- An emergent gap is also very detrimental to weld strength and spot welding can hinder emergent gaps in some degree.

Abstract

In this paper, we studied the femtosecond laser welding of soda–lime–silicate glass and a pure aluminium film by employing a 100 kHz femtosecond laser. The morphology of the whole welding region is the same as that of the glass. The sizes of the defects in the welding regions must be very small, so that it looks like that the welding region integrates seamlessly with the glass. However, the real profile of the welding region can be still plotted after analyzing the SEM-EDS data and the Raman data. The welding region can be divided into three major parts: the area formed from the solidifying plasma of aluminium, from the solidifying mixed plasma of two materials, and where the silicate networks are partly destroyed, respectively. This means that the material properties in the welding region gradually transform from ductile to brittle towards the beam. We also studied two types of welding, continuous scanning welding and spot welding, and the lap shear strengths are compared in the different total areas of the welds. The data show that both types of the laser welding can provide strong strength. We attribute the high strength to the density of the welding region which results from a step welding dynamics caused by a relatively low repetition rate and a relatively high intensity.

Keywords

Fs laser welding; Continuous scanning welding; Spot welding; Shear strength; Shock wave; Plasma

5. Results of Raman spectra in the welded...
6. Lap shear strengths by continuous scan...
7. Discussion about the possible reasons f...
8. Conclusions

CRediT authorship contribution statement

Declaration of competing interest

Acknowledgement

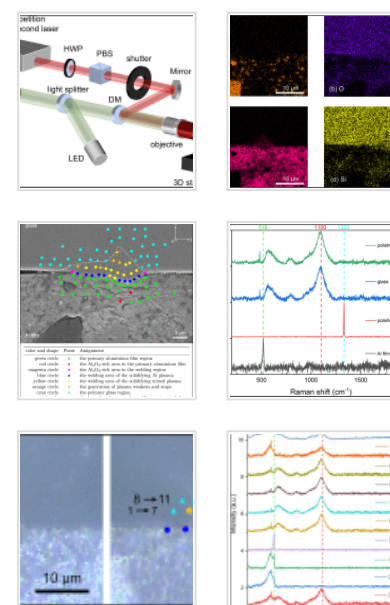
Appendix A. Supplementary data

Data availability

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1. Introduction

Ultrashort pulse laser welding has been widely investigated in recent years [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14]. The welding parameter sets are very wide and the reported welding strengths are also disperse, whatever glass to glass or glass to other materials. There are several excellent review papers about the welding of glass-to-glass [15], [16], [17] and glass-to-other materials [18], [19], [20].

In the experimental research of ultrashort pulse laser welding of glass and aluminium (pure or alloy), the results are also quite divergent. The dissipation of heat in glass is generally less than 10 μ s [21], [22]. The order of hundreds of kHz and a relatively low pulse energy should be better since metal has much higher thermal conductivity than glass [23]. However, several works showed that the shear strengths are high when using high pulse energy and 1 kHz femtosecond laser [9], [11]. The large negative defocusing was very usual welding condition [23]. The geometric focus at the interface was also reported [12]. All these divergences show the complications of the welding dynamics which are sensitive to both the laser parameter sets and the material properties. The researchers also began to focus on the characterisation of weld zone reactions [10], [11] and mechanism of the welding process [12]. The formation of nanocrystalline silicon reactions and the transitional phases (γ - and δ -Al₂O₃) was reported [10], [11].

The nonlinear propagation of the ultrashort pulse inside material has been thoroughly investigated [24], [25]. Welding glass and silicon using the filament phenomenon was first reported in 2006 [26]. The effect of self-focusing on welding was recently researched [6], [27]. So far the simulation work most relevant to the femtosecond (fs) laser welding has focused on the evolution of the temperature distribution and the free electron density [15], [28]. However, many simulation studies about the ablation process [29], [30], [31] and the shock wave dynamics [32] are helpful for understanding the manifold experimental phenomena in the ultrashort pulse laser welding.

In this paper, we studied the femtosecond laser welding of soda–lime–silicate glass and a pure aluminium film. The SEM photograph shows that the morphology of the welding

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Tables (1)

 Table 1. The average width of the welds ...

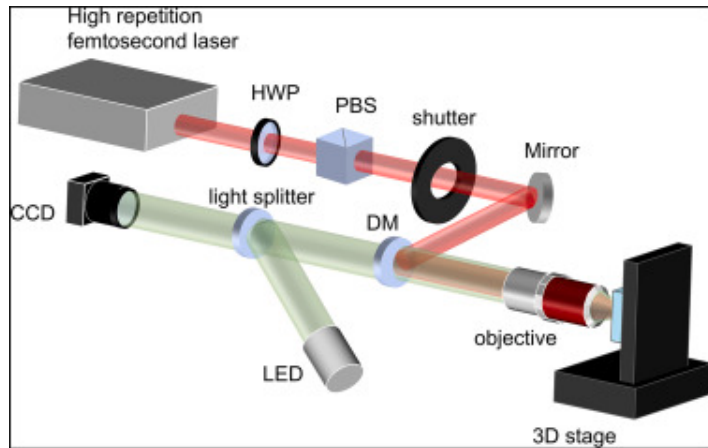
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region looks the same as that of the glass. The SEM-EDS data and the Raman curves measured at the welding region point by point illustrate the profile, the height, and the rough distribution of the compositions in the welding region. The lap shear strengths of two types of fs laser welding, continuous scanning welding and spot welding, are measured and compared. In the case of the continuous scanning welding, the robustness of a single weld with the length of 8 mm is very low and the maximum load at fracture is only 2 N. It is interesting that both the load at fracture and the robustness in the case of two welds abruptly increase. For the spot welding, the load at fracture and the robustness are both sufficient when the total area of the weld spots is the same as that of a single weld. We discussed the possible reasons for this strong shear strength.

2. Experimental setup, materials, and methods

The fs laser welding system is shown in Fig. 1. A commercial fiber femtosecond laser (BFL-1030-20L, BWT Beijing Ltd.) with a repetition rate of 100 kHz, a wavelength of 1030 nm, and a pulse duration of 350 fs was used. The $1/e^2$ diameter of the laser beam was about 3 mm. The N.A. of the objective lens is 0.4, and the working distance is 20 mm. Accordingly, the $1/e^2$ diameter of the focal spot is about **4.39 μm** . The glass was mounted on the aluminium film and both were fixed by the fixture. The sample to be welded was placed on a three-dimensional (3D) stage. The laser beam passed through the glass and was vertically incident on the interface of two materials. The welding process was accomplished by controlling the trajectory of the 3D stage and the on/off state of a shutter. A CCD observation system was employed for real-time monitoring of the welding process.



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Fig. 1. The fs laser welding set-up. The laser power for welding can be adjusted by the half-wave plate (HWP) before the polarization beam splitter (PBS).

The glass used in the experiments is an ordinary soda–lime–silicate glass, which is brittle and not heat-resistant. It has a refractive index of about 1.5, and its original size is about **75 mm × 25 mm × 1 mm**. The aluminium film with a purity in excess of 99.95 % is used for the experiments. The thickness of the aluminium film is **20 μm**. The surface roughness (Ra) is about 300–500 nm and the optical contact cannot be obtained (see Fig. S1, Supporting information).

The film is between two glass slides and a clamping device is used for holding the glass and film. The clamping device is on a 2D adjustable holder which makes the interface perpendicular to the laser beam. There is no gravitational force upon the aluminium film because the laser beam is parallel to the optical platform. The force is not accurate in each experiment but it is not strong in order to avoid inducing high stress on the above glass slide. The scheme of clamp is shown in Fig. S2 (Supporting information). The low power laser beam is focused on the interface as the reference position of the zero defocusing amount. The scanning strategy is a grating pattern without corners.

The cross-section of the weld, which is perpendicular to the laser scanning direction, was prepared by grinding and polishing. After polishing, the welding samples were placed in the acetone solution and cleaned for about 6 min in an ultrasonic wave cleaner. A micro-Raman spectrometer (Renishaw, inVia microscope) and a scanning electron microscope (ZEISS, Sigma 500) were used to characterise the element distributions and the molecular bonds inside and around the welding region on the cross-section. The shear strength was measured using a universal material testing machine (LSplus, Ametek). The tensile speed was set to 5 mm/min. The scheme of test is shown in Fig. S3 (Supporting information).

3. Parameter sets for a successful welding

3.1. The width of the welds at the different parameter sets for a successful welding

A successful welding is defined by following two conditions. One is that there is no crack within the glass region of the welding sample. The welding strength is also strong at many parameter sets even though there are cracks in the glass. Another is that the welding sample (40 welds with a length of 8 mm) does not fracture after being pulled by hands. Many parameter sets can be used for obtaining a successful welding and parts (scanning velocity is 0.6 mm/s) are shown in Table 1, in which d is the geometric defocusing amount at the low power. We measured some samples and the fracture points are all in the part of the aluminium film. The welds act as the clamp and the real shear strength of the welds cannot be obtained from testing these samples (see Fig. S4, Supporting information).

Table 1. The average width of the welds under optical microscopy at different powers and defocusing amounts d when the scan velocity is 0.6 mm/s. The units of the width and d are both μm . The width always decreases (about 18–20 μm) when the bilateral glass areas melt. P is the laser power. $\uparrow/\downarrow$ means that the results are obtained at a slightly higher/lower velocity. The deductions labeled by color boxes will be discussed in the following section.

P (mW)	$d = 0$	$d = 15$	$d = 30$	$d = 40$	$d = 50$	$d = -15$	$d = -30$	$d = -50$
100	11.5	fail	12.7↑	fail	fail	13.8↓	fail	fail
200	16.0	15.5	15.5	14.1↑	fail	18.2	fail	18.8↓
300	16.4	17.1	17.7	17.2	fail	20.4	fail	21.5
400	18.7	19.3	19.3	17.7	fail	21.0	20.0	23.8
500	21.5	20.4	20.4	20.2	fail	22.6	22.6	24.9
600	crack	21.5	22.1	19.2	fail	crack	24.0	27.6
700	–	melt	22.1	melt	fail	–	24.0	27.6
800	–	melt	22.6	melt	fail	–	crack	28.2
900	–	melt	22.8	melt	19.7↑	–	–	28.7
1000	–	melt	melt	melt	21.8	–	–	crack
1200	–	crack	melt	melt	22.8	–	–	–
1500	–	–	melt	melt	24.8	–	–	–

 usual welding
 filament welding
 far from filament
 ablation
 low density

There are three points that need to be mentioned before discussing the data in Table 1. First is about the velocity in the case of the lowest power for a successful welding. The down-arrow ↓ in Table 1 means that the velocity must be a little less than 0.6 mm/s for a successful welding. This implies that more pulse numbers per point N are needed. The up-arrow ↑ in Table 1 indicates that N needs to be reduced.

Second is about the melt phenomena. The minimum power for melting the bilateral glass areas along the weld (the scanning direction) is 700 mW. It needs to be noted that the melt of these glass areas only randomly takes place. Sometimes the molten areas are invisible under the optical microscopy even though the power increases and the other parameters remain the same. Obviously, the melt depends not only on the power but also on the intensity (see the column of $d = 50 \mu\text{m}$). The width always decreases slightly when this melt occurs. In addition, this molten zone does not bond to the film. Thus, the occurrence of these bilateral heat affected zones only introduces the defects in the glass.

Third is about the self-focusing and the filamentation. We calculated the focus shifts caused by self-focusing at the different P/P_{cr} and the different defocusing amounts by use of two different methods and the relative discussions are shown in Sections S5–S7 (Supporting information). In our experiments, a very thin highly transparent modification region appears under the optical microscope when the beam is directly focused inside the

glass and the power exceeds 250 mW. The filamentation becomes stable beyond this value.

3.2. $d = -50 \mu\text{m}$: a usual welding mode in the case of the negative defocusing

The self-focusing can be surely inhibited in the case of a large negative defocusing amount. Combining the results in Table S2 and the discussions in Sections S5–S7 (Supporting information), we infer that the welding at $d = -50 \mu\text{m}$ is a usual welding mode and the focus is outside the glass. The failure at 100 mW should be due to low intensity. The demand for a slower velocity illustrates that the intensity at the interface is not sufficient at 200 mW. The collapse position will move towards the interface when the power increases. The crack at 1000 mW may arise from that the collapse position is too close to the interface (the focus may be inside the glass).

3.3. $d = -30 \mu\text{m}$: the collapse positions pass the interface

The power range for a successful welding must abruptly narrow when the collapse position passes the interface. According to the experimental results, we can determine this critical defocusing amount. Here it is $d = -30 \mu\text{m}$. The record of d is practical, even though it is hard to determine the exact position of $d = 0$ when the beam is focused inside a medium [33]. Depending on d , we know that it is a usual welding or a filament welding at a certain parameter set.

At this critical defocusing amount, the intensity at the interface is too low to weld at 100 mW since the focus should be outside the glass (see Table S2, Supporting information). And the welding fails at 300 mW mostly because the ablation becomes dominant when the collapse position is very close to or passes the interface. The focus position should be inside the glass when the power increases to 400 mW. In this case, it becomes the filament welding mode. The collapse position moves towards the beam with the increase of the power, however the filament length consequently augments. This is why the power range for a successful welding is so narrow.

3.4. $d > -15 \mu\text{m}$: different welding mechanisms at the different powers

We tend to consider that the focus is outside the glass at $d = -15 \mu\text{m}$ and 100 mW because the velocity must reduce slightly for a successful welding. This deduction corresponds to the results in Table S2 (see the discussions in Supporting information). It is hard to determine what the welding mode is at 200 mW. The focus shift changes so nonlinearly with the power and the defocusing amount. According to the permanent damage threshold of 250 mW, we think that the self-focusing inside the glass should take place at 200 mW and $d = -15 \mu\text{m}$. It is possible that the focus is inside the glass but the abrupt collapse does not occur because other nonlinear interactions can balance the collapse at such a low power. It seems that the filamentation in the glass must play a key role in the welding at 300–500 mW, according to the data in Supporting information.

3.5. $d > 0$: filament welding is dominant

The filament welding becomes the dominant welding dynamic in the most parameter sets in the case of $d > 0$. The nonlinear propagation can be stable within a power range of several P_{cr} [25]. Much more power is needed when $d = 50 \mu\text{m}$ because the interface is too far from the filament. The length of the filament increases with the power and the intensity at the interface finally becomes appropriate when the power increases to 900 mW. The melt never occurs until the power is excessively high and the glass cracks since the absorption is low in the filament. The welding results in Table 1 show that the stable and successful welding can be achieved in the filament welding mode.

The welding results show very irregular at 100 mW at $d > 0$. We tend to consider that it is always a usual welding at 100 mW. The fluctuating results arise from the effect of both the intensity and the pulse number per point N . This corresponds to the value of n_2 of soda-lime glass (see Supporting information). It is not certain about the welding mode at 200 mW.

3.6. The saturation of the width of the welds

From Table 1, it is easy to misread that the reason for the saturation of the width of the welds arises from the filamentation. In fact, the widths of the welds are exactly the same in the case of the welding of glass and graphite in the usual welding mode. In addition, the width actually becomes a little narrower when the melt occurs in the bilateral glass areas at a higher power. These results demonstrate that the most dominant factor influencing the width of the welds is the interaction of the fs laser and materials in our experiment, rather than the beam spot, the intensity, or the power. This shows that the dynamics of the single pulse are most important rather than the heat accumulation effect of the multiple pulses when welding by employing a 100 kHz fs laser. It is why the intensity at the interface is always the most significant factor in our welding experiment.

3.7. Summary of the parameter sets for a successful welding

A successful crack-free welding can be achieved with the parameter sets of both the usual welding and the filament welding. The length of a filament mainly depends on the pulse-peak power. However, the focus shift also depends on d . In summary, the power determines the filament length and d controls the distance between the filament and the interface. Both the power and the defocusing amount determine the intensity at the interface.

4. Results of SEM-EDS in the welding region of the filament welding sample

4.1. The morphology of the SEM photograph of the filament welding sample

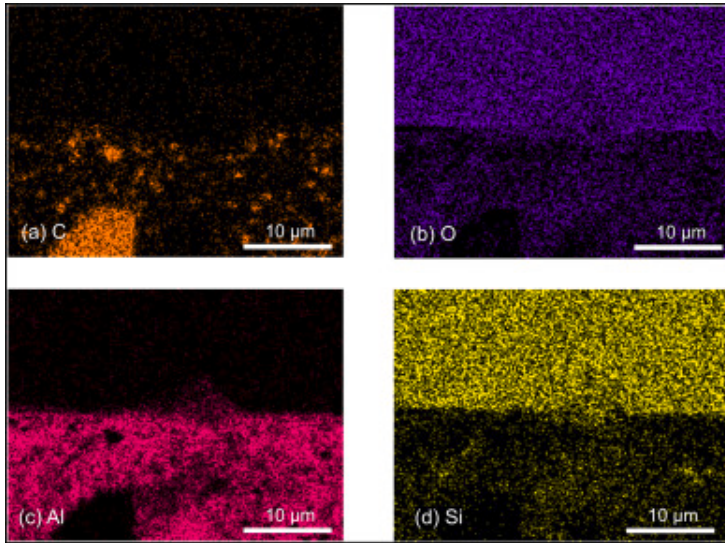
We measured the SEM photographs and maps of the filament welding samples at three typical parameter sets. Their morphologies look exactly the same. Only the boundaries between the welding region and the aluminium film are clear. The boundaries between the welding region and the glass region cannot be discriminated directly from the SEM photograph. It seems that the welding region integrates seamlessly with the glass (see Fig. 3). There are no pores, microcracks, or cavities as reported by other references [10],

[11]. In addition, the obvious damages in the glass (we looked for them at a large magnification) cannot be found above the welding region in several tens of micrometer range, even though the filament must exist.

In the SEM maps, only one sample can show an obscure profile of the welding region. This sample is made at $d = 15 \mu\text{m}$ and 400 mW. The samples welded at this parameter set are used in the following SEM-EDS data and the Raman spectra because a rough measurement range is known.

4.2. The SEM maps of the welding sample

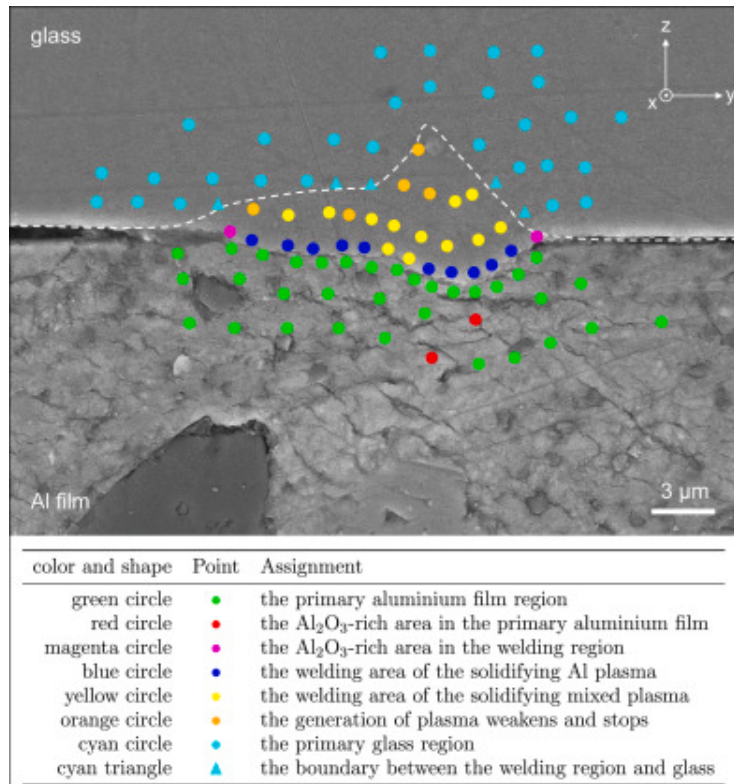
Fig. 2(a)–(d) show the maps of four elements in the welding region, carbon, oxygen, aluminium, and silicon (i.e., C, O, Al, and Si), respectively. As Fig. 2(a) shows, the distribution of the carbon atoms in the aluminium film region is quite nonuniform because the diamond nanoparticles in the polishing paper cannot be completely removed. A great amount of the oxygen atoms randomly occur in the aluminium film region as shown in Fig. 2(b). The profile of the welding region can be vaguely recognized only from the map of the aluminium atoms (see Fig. 2(c)). In the map of the silicon atoms, the profile is very obscure. The dispersed silicon atoms in the aluminium film region may arise from the debris of the glass produced during the polishing process. It is interesting that this not very clear profile just matches the same dim profile observed under the optical microscope. However, they are both not the real profile of the welding region.



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Fig. 2. The maps of four elements: (a) carbon, (b) oxygen, (c) aluminium, and (d) silicon.



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Fig. 3. The SEM photograph and the different areas inside and around the welding region discriminated by use of the SEM-EDS data. The x axis is in the scanning direction and the z axis is towards the laser beam.

4.3. The SEM-EDS data and the real profile of the welding region

The concentrations of all possible elements at the positions inside and around the welding region are measured point by point by SEM-EDS, as shown in Fig. 3 and Tables S3–S11 in Supporting information. The concentration of the carbon atoms is extraordinarily as high as 30–67 at% and its influence must be removed before analysis. After removing the contribution of the carbon atoms and normalizing the concentrations

of other elements, the concentration of the aluminium atoms is maintained at 0.6–0.9 at% in the glass region far away from the welding region. The concentration of the silicon atoms is about 24 at% in the primary glass. The concentration of the oxygen atoms is about 67 at% and the ratio of the atom number of oxygen to the total atom number of silicon, sodium, magnesium, and calcium is about 1.5, which corresponds to the usual value of soda–lime–silicate glass. In the primary aluminium film region, the concentration of the silicon atoms decreases to 1–9 at%. The concentration of the aluminium atoms obviously fluctuates with the concentration of the oxygen atoms. Nevertheless, there is a feature in the aluminium film region where the concentration of the oxygen atoms is much lower than 67 at% except for one instance. This ratio is possibly near 1.5 when meeting the Al_2O_3 -rich area in the aluminium film region. Even so, the concentration of the silicon atoms must be very small in this area. According to the ratios of the concentrations of these elements, the welding region and the primary material region can be separated and the different welding areas with distinct features can also be discriminated. The results are shown in Fig. 3.

At the green circle points, the concentration of the aluminium atoms is high and that of the silicon atoms is low. The ratio of the oxygen atoms to the aluminium atoms is much lower than 1.5. The morphology of this region is evidently the same as that of the primary aluminium film and is obviously different from that of the glass. So, they are absolutely in the primary aluminium film region. The red circle points are just in the random Al_2O_3 -rich area in the aluminium film since the concentration of the silicon atoms is low and the ratio of the oxygen atoms to the aluminium atoms is exactly 1.5. The morphology at the red circle points also shows moreover that they are in the aluminium film region.

The most interesting thing is the composition at the blue circle points. The concentration of the silicon atoms is low and that of the aluminium atoms is high. It is easy to be misjudged as the aluminium film region if only looking at data. To observe the morphology of this region in Fig. 3, it is so different from that in the primary aluminium film region and exactly the same as that in the primary glass region. They are certainly in the welding area formed from the solidifying plasma of aluminium. On account of the

same morphologic reason, the magenta circle points are in the Al_2O_3 -rich area in the welding region.

The concentrations of the aluminium atoms and the oxygen atoms are both more than 10 at% at the yellow points, so they are certainly in the region formed from the mixed plasma of two materials due to both the known dynamics of an fs laser welding and the following Raman data. The concentration of the silicon atoms is 20–22 at% and that of the oxygen atoms is 58–62 at% at the orange circle points. The composition looks similar to that of glass. However, the concentration of the aluminium atoms is 3.76–9 at%, which is obviously higher than that in the primary glass. These points should be in the welding region where the movement of the plasma slows down and the growth rate weakens.

The composition at the cyan triangle points is almost the same as that of glass, but the concentration of the aluminium atoms is 1–2 at%, which is a little higher than the value in the primary glass. These points may be in the primary glass due to the limited accuracy of the SEM-EDS data. It is also possible that they are located in or very close to the boundary between the welding region and the glass. A very few aluminium atoms enter the glass region by thermal diffusion. The cyan circle points must be in the primary glass because the composition is almost identical. The above analysis is summarized in Fig. 3.

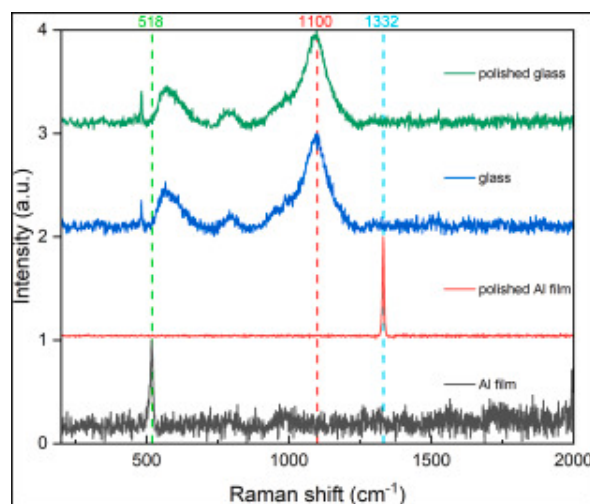
4.4. Summary of the SEM data

According to the above analysis of the SEM-EDS data, the height between the bottom of the welding region and the primary interface is about $2\text{ }\mu\text{m}$, and the maximum height of the whole welding region is about $6.7\text{ }\mu\text{m}$. The maximum width of the welding region is about $16\text{ }\mu\text{m}$, which is $3\text{ }\mu\text{m}$ smaller than the average width under the optical microscope ($19.3\text{ }\mu\text{m}$). The concentration of aluminium in the welding region gradually reduces towards the beam. The whole welding region has a high density and the size of defects must be very small.

5. Results of Raman spectra in the welded region

5.1. The Raman spectra of the primary materials

Fig. 4 shows the Raman curves of the primary aluminium film and the glass before and after polishing. There are three typical bands in the Raman curve of soda–lime–silicate glass, 570 cm^{-1} , 787 cm^{-1} , and 1100 cm^{-1} , respectively [34]. The band with a peak at 1100 cm^{-1} results from tectosilicates. It indicates that there are vibrations of the bridging oxygen in the Si–O–Si linkage [34]. The existence of this band means that there are the silicate networks. The polishing has no effect on the Raman curve of the glass. Metal has no Raman response at all. The sharp peak at 518 cm^{-1} in the Raman curve of the pure aluminium film may arise from Al_2O_3 (its shape suggests that it is not noise peak). This peak occurs very randomly and this corresponds with the random red circular points in Fig. 3. After polishing, a new peak at 1332 cm^{-1} occurs, which arises from the residual diamond nanoparticles. The glass is rigid, so the residual diamond nanoparticles can be removed after the ultrasonic cleaning. However, the aluminium film is too soft to remove them completely.



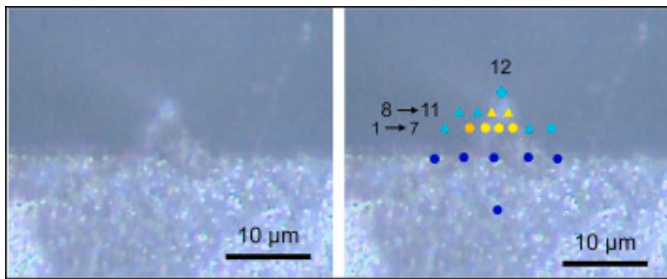
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Fig. 4. The Raman curves of primary glass and aluminium film before and after polishing.

5.2. The Raman spectra of the welded sample

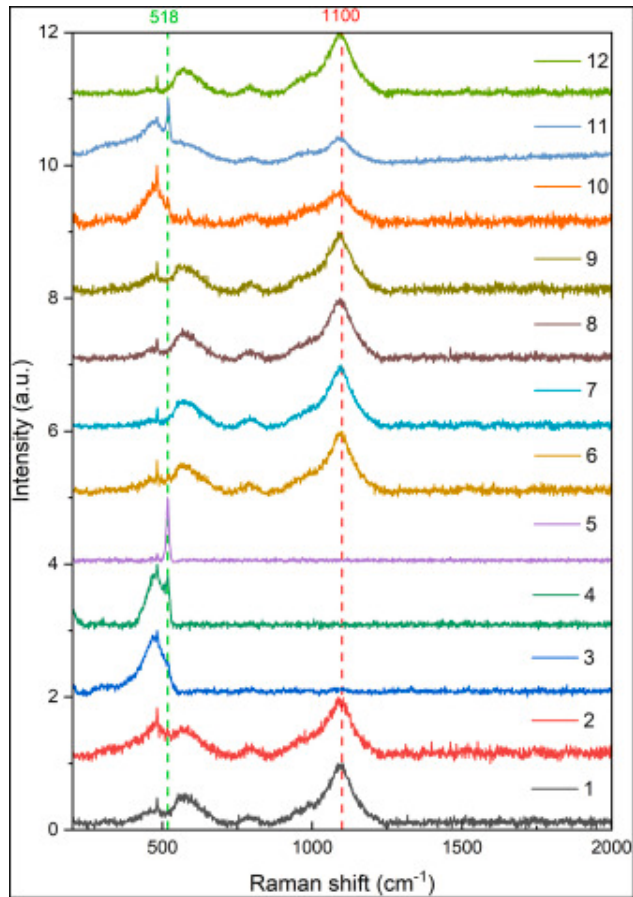
The sample for Raman spectra is not the same sample used for SEM. In addition, the spatial resolution of a usual Raman spectrometer is much lower than that of SEM-EDS. However, the following analysis is not influenced because the relative positions among these areas do not change. The photograph of the sample under the optical microscope in the Raman spectrometer and the positions at which the Raman curves are measured are respectively marked in Fig. 5. The colors and the point types have the same meanings as those in Fig. 3. The Raman curves are shown in Fig. 6. Curves 1–7 are the Raman curves measured in the plane about $3\text{ }\mu\text{m}$ above the primary interface. Curves 8–11 and 12 are the Raman curves in the planes $4.5\text{ }\mu\text{m}$ and $6\text{ }\mu\text{m}$ above the interface respectively.



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Fig. 5. The photograph of the welding sample under the optical microscope in the Raman spectrometer and the positions at which the Raman curves are measured. The number corresponds to the same number of the Raman curve.



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Fig. 6. The Raman curves measured in the welding region.

There are no characteristic bands of soda–lime–silicate glass in the Raman curves at the root of the welding region (blue circle points). Only the sharp peak of diamond at 1332 cm^{-1} emerges in these Raman curves. The peak assigned to Al_2O_3 occurs only once. The Raman curves at the blue points are only another evidence of the welding area formed from the plasma of aluminium, so we skip them (not shown here).

5.3. The Raman spectra in the mixed plasma region

The three bands of the glass (the peaks at 570 cm^{-1} , 787 cm^{-1} , and 1100 cm^{-1}) all disappear in Curves 3–5. This means that there are no silicate networks. This is very different from the Raman spectra in the modified region reported in Ref. [16]. The characteristic bands are mostly below 500 cm^{-1} in Curves 3 and 4. All possible oxides have some characteristic peaks below 500 cm^{-1} , for example, microcrystalline silica [35], CaO [36], and $\alpha\text{-Al}_2\text{O}_3$ [37]. In addition, the down-shifted broad band of the amorphous silicon is also in this range [38]. The optical phonon of crystalline silicon has a characteristic band at 520 cm^{-1} and this characteristic peak will red shift due to the small size of crystalline silicon particles [38]. The existence of the nanocrystalline silicon in the welding region was first reported in Ref. [10]. It results from the chemical reaction of SiO_2 and Al [10]. Besides, the existence of $\gamma\text{-Al}_2\text{O}_3$ and $\eta\text{-Al}_2\text{O}_3$ reported in Ref. [10] cannot be excluded because they do not exhibit any bands in the Raman spectra [37]. Curve 5 looks exactly the same as the Raman curve of the primary aluminium film, in which there is only a peak at about 518 cm^{-1} due to Al_2O_3 . It should be attributed to the fracture of Al_2O_3 in the primary aluminium film. We do not think that this peak is assigned to crystalline silicon because the width is too narrow.

Indeed, it is impossible to know the exact compositions in this area only from the Raman spectra. It is at least certain that the positions where Curve 3–5 are measured definitely correspond to the yellow points in Fig. 3. These curves demonstrate that the welding region formed through the direct solidification of mixed plasma of two materials, rather than aluminium atoms diffusing into the thermally molten glass. The generation of the plasma and subsequent direct solidification must take place in an ultrashort time, therefore this hinders the recovery of the silicate networks. Curve 3–5 are so different even though they are measured at the adjacent positions. These also show a feature of chaos, which is far from the Raman data in the modification region in the glass [16]. This leads to the totally heterogeneous microstructures in three adjacent areas at the same height.

5.4. The Raman spectra of the mixed plasma decay region

In Curves 10 and 11, the peak of the oxides below 500 cm^{-1} exceeds the peak at 1100 cm^{-1} . This means that the silicate networks of glass must be fractured to some degree. The shape of the bands below 500 cm^{-1} in Curves 10 and 11 is similar to those in Curves 3 and 4. This is a feature of the generation of the plasma and the modification region in the glass–glass welding [16]. Obviously, the mixture of the different plasmas weakens, so that a part of silicate networks can remain intact. In Fig. 5, these positions are marked by the yellow triangular points for indicating that they may be on the boundary of the welding area formed from the mixed plasma.

In Curve 2, the high band at 1100 cm^{-1} emerges, however its relative amplitude visibly decreases when compared with the Raman spectra of the glass. The low-frequency bands visibly expand into the range below 500 cm^{-1} . This area corresponds to the orange points in Fig. 3. Only a few silicate networks of the glass are fractured. The generation of the plasma tends to stop.

5.5. The Raman spectra at the boundary of the welding region

The bands below 500 cm^{-1} still exist in Curve 1. The total area below 500 cm^{-1} strongly shrinks and the shape of Curve 1 becomes very similar to the Raman curve of the glass. Curves 6, 8, and 9 are almost the same as Curve 1. They may be the Raman spectra at the cyan triangle points in Fig. 3, which is just on the boundary between the welding region and the glass. We think that there is a very thin molten glass region at the boundary from these Raman curves because they are similar to the Raman spectra in the heat-affected zone of the glass–glass welding sample [16]. Curves 7 and 12 are absolutely the same as the Raman curve of the glass and these areas correspond to the cyan circle points in Fig. 3.

5.6. Summary of Raman spectra in the welding region

In summary, Raman spectra show that the compositions in the welding region are very complicated. The features of the matter in the welding region approximately gradually

alter towards the laser beam even though their morphology looks exactly the same in SEM photograph. The silicate networks also recover little by little. The boundary layer of the welding region and the primary glass seems very thin. From Raman spectra, it is clear that the welding areas formed from the solidifying plasma are dominant in the welding region.

According to the SEM data (Fig. 3, S8 in Supporting information) and the Raman data (Fig. 6), the major bonding mechanism must be chemical bonding, rather than mechanical interlocking. However, there is not a simple way to confirm the exact chemical bonds in the welding region. It is partly because the compositions in the mixed plasma region are multiple and they strongly change with positions in the core area.

6. Lap shear strengths by continuous scanning welding and spot welding

6.1. Welding parameter sets of continuous scanning welding and spot welding

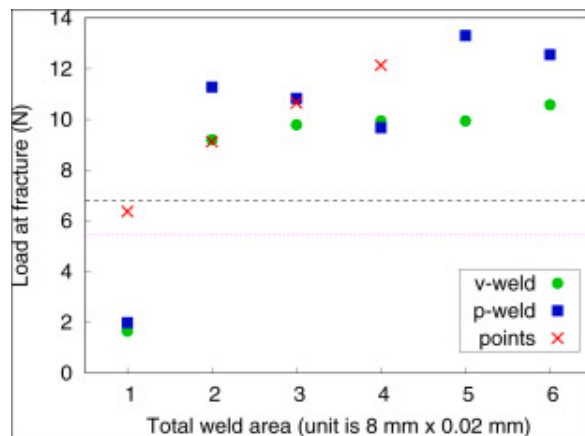
For obtaining the weld strength, the lap shear strengths of 1–6 welds with a length of 8 mm were measured. The laser power increases to 500 mW in order to compare the results of continuous scanning welding and spot welding. The scanning velocity is 1.2 mm/s and $d = 15 \mu\text{m}$. The width measured by the optical microscope is always about $3 \mu\text{m}$ larger than the maximum width measured by SEM. Thus, the width of a single weld used for calculating the strength is $20 \mu\text{m}$ when it is about $22 \mu\text{m}$ under the optical microscope. Hence, the total weld area of a single weld with a length of 8 mm is accordingly 80 % of the cross-sectional area of the primary aluminium film with a width of 10 mm and a thickness of $20 \mu\text{m}$.

In the experiments of the spot welding, the opening time was not optimized carefully and only three values were tested. The shear strength tested by hand is much better when the opening time of the shutter is 16 ms compared to those in the cases of 8 ms and 32 ms. Hence, the pulse number per point is 1600. This value is not very exact due to the

operating principle of the mechanical shutter. The other parameters are the same as those used in the scanning welding. The diameter of a single weld point is **22 μm** under the optical microscope, which is the exactly equal to the width of a single weld in the scanning welding. Hence, the real diameter is estimated to be **20 μm** . Depending on this value, we can calculate the number of the weld points when the total weld area is equivalent to the weld area of a single weld with a length of 8 mm. It needs 400 points. In spot welding, there is a square array with a weld point distance of **100 μm** , i.e., **20 $\times$ 20** points array.

6.2. The results of the shear strength

The green and blue points in Fig. 7 represent the load at fracture of the welds which are perpendicular and parallel to the direction of the load respectively. The red $\times$ -shaped points are the results of the spot welding. The unit of the weld area is **8 mm $\times$ 20 μm** , which is just the weld area of a single weld.



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Fig. 7. The loads at fracture change with the total weld area in the case of continuous scanning welding and spot welding. The area of the single weld is **8 mm $\times$ 20 μm** (or **8 mm $\times$ 0.02 mm**).

For the aluminium film, the yield strength is 34 MPa and the fracture strength is 103 MPa (both are provided by the supplier). The load calculated using the yield strength of the aluminium film with a width of 10 mm and a thickness of **20 μm** is 6.8 N, which is labeled by the horizontal dashed line in Fig. 7. The load calculated using the yield strength of the aluminium film and the area of a single weld is 5.44 N, which is labeled by the horizontal magenta dotted line in Fig. 7.

The position of the fracture point can indicate whether the data can be used for calculating the lap shear strength. The fracture point must be in the part of the aluminium film if the load exceeds this dashed line. In addition, the fracture point is always in the part of the aluminium film when the load exceeds the magenta dotted line. In these two cases the results cannot give the real shear strength of the welds because the bonding is high and the welds act like the fixture.

6.3. The shear strength of the scanning welding

It is so obvious that the load at fracture of a single weld is very poor as it is less than 2 N. In fact, the joint fails many times and these two samples are selected. The fracture takes place just in the welding region. The fracture sections are random, i.e., sometimes in the glass region near the weld and sometimes in the aluminium film below the weld. Only in this case, the shear strength can be obtained and they are 10.32 MPa and 12.41 MPa, respectively.

However, the load of the sample with two welds abruptly increases and the load at fracture jumps over the gray dashed line (9.2 N and 11.28 N, respectively). The fracture occurs in the part of the aluminium film and the fracture point is close to the welding region. The lap shear strength is 28.8 MPa and 35.2 MPa when using the total welding area. The fracture strength is 45.6 MPa and 53.7 MPa when using the cross-section at fracture given by the device. The robustness also increases and the welding never fails.

The loads at fracture fluctuate when the number of the welds is more than three and the fracture points move to the middle of the aluminium film. Logically, it is impossible for

the lap shear strength to linearly drop with the weld area, so these strengths, which can be calculated from the experimental data, are only the fracture strengths of the aluminium film. The welds play a role as one fixture in the tensile test, however these welds cannot provide a uniform harmless clamping condition. These variable test conditions cause the fluctuation of the load for 2–6 welds in Fig. 7. Accordingly these loads at fracture are much lower than 20.6 N, which is the value obtained in a uniform clamping condition.

6.4. The shear strength of the spot welding

It is an amazing result that the maximum load at fracture can increase to 6.37 N when the total weld area of the point array is just equivalent to the area of a single weld. The fracture point takes place in the part of the aluminium film, just like the phenomenon in the previous case of 2–6 welds. The lap shear strength is 50.7 MPa when using the total welding area. According to the above discussion, the fracture strength is 35 MPa when using the cross-section at fracture given by the device. Only one thing is definite that the lap shear strength can exceed the yield strength of the aluminium film in the best results.

In addition, the robustness of the welding becomes quite strong and the failure of the welding never occurs. Three samples with one unit area were made. The fracture points of other two samples occur at the welding region, so that the shear strengths can be obtained. They are both high, 36.1 MPa and 32.3 MPa, respectively. The error bar is not meaningful for our data because the film is too thin and the welding strength is too high.

Further increasing the total area of the point array, the fracture points behave as in the previous cases of 2–6 welds. Only the increase in the load at fracture with the weld area shows a monotonic change.

6.5. Summary of the shear strength

It is hard to say that the spot welding is better than the continuous scanning welding because the strength of the sample with two welds is also strong, especially considering

the poor strength and the poor robustness of a single weld. The first weld seems to provide some bonding force, and with its help the second scanning can perform well. However, the results certainly show that the spot welding is a good choice for a strong joint. We will discuss some possible reasons for these strong strengths in the next section.

7. Discussion about the possible reasons for the strong shear strength

7.1. The effects of the characteristics of the welding region

The morphology of the welding region is obviously different from those reported in many studies at the similar magnification [10], [11], [12]. There are no pores, cavities, or microcracks. There is no keyhole in the aluminium film region even though the generation of the plasma must exist. There is also no visible boundary between the welding region and the glass region. The above factors all support a high strength because the sizes of the defects in the welding region are truly very small.

Ref. [10] reported that there is a thin interface layer consisting of Al_2O_3 above the boundary of the primary aluminium and the morphology of this thin layer is visibly different from the glass. The Al_2O_3 layer is also brittle due to its covalent bonds. For our sample, most in the same position is a layer formed from the solidification of the aluminium plasma. The size of the grains in this layer must be very fine so that its morphology in SEM looks like the glass, rather than the aluminium film. In the view of the material properties, we think that this aluminium layer is a beneficial factor. This means that the ductile material gradually transforms into the brittle material in the welding region.

At 1030 nm, two major possible materials can linearly absorb the light. They are aluminium and the silicon nanoparticles. The silicon nanoparticles arise from the reaction of Si_2O_3 and Al [10]. According to the SEM-EDS data, the vertical distribution of the concentration of aluminium atoms in the welding region shows a coarsely gradual feature along the laser beam direction. It seems that the linear absorption layer moves (grows)

towards the laser beam piece by piece with a decaying absorbance. As Raman spectra show, the silicate network also recovers in a similar manner. This also demonstrates the gradual variation of the material properties in the welding region. We will discuss the possible reasons in the next text.

7.2. High pulse energy density deposits at the interface due to the filamentation (the nonlinear propagation) in the glass

A large negative defocusing amount can avoid self-focusing, however, it will strongly decrease the intensity at the interface because the state is very unstable if the interface is too close to the collapse position. The higher repetition rate becomes essential if the intensity at the interface is too low. The laser with higher repetition rate always performs well for melting the glass. However, the dominant welding dynamics also transform from the plasma generation caused by a single pulse to heating due to thermal accumulation of multiple pulses. Furthermore, there should be a large size of the molten pool. The convection and turbulence will destroy the advantages of the gradual features above-mentioned when a fluid phase exists in a large volume.

The filament in a transparent medium is a special propagation mode and the nonlinear absorption can remain small within the filament. As a result, the filament acts like a nonlinear lens which can focus the beam at the interface with an appropriate intensity. In addition, the filament can also function as an adjustable lens. The appropriate intensity at the interface can be controlled by adjusting the positive defocusing amount. Finally, the plasma generated by each pulse can be regulated.

7.3. Heat and cool cycle: a step-by-step process of the welding dynamics

It is better that the size of the modification region (the welding region) increases step by step if requiring a gradual variation of the material properties. The permanent heat molten pool during the welding needs to be impeded to avoid the convection and turbulence of the fluid. A welding by the fs laser with a relatively low repetition rate can satisfy this. The temperature caused by the previous pulse drops below the softening

temperature and/or the melting temperature before the next pulse arrives. The pulse always meets the gradual modification region and the dynamics of the fluid (convection and turbulence) can be extremely decreased because of the limited volume of the instantaneous molten pool. The concentrations of the materials which can linearly absorb the light decrease with the height (towards the beam), so that the material properties in the welding region gradually change. The chaos of the compositions in the mixed plasma area and the asymmetry of the profile of the welding region are most likely due to the shock wave. The process above-mentioned requires sufficient intensity at the interface.

The morphology of the welding region appears to have the same morphology even though the bilateral glass areas melt at 800 mW. This may result from the fact that heat accumulation requires a large pulse number. We think that the step process of the welding dynamics caused by heat and cool cycle is most dominant in our experiments in the case of $d > 0$, where the filament welding is dominant. However, the melt in the bilateral glass areas is a detrimental factor because it is only the source of the defects in the glass and no any contribution to the strength.

7.4. Shock wave: repeated laser peening

The fs pulse with 1 J/cm^2 is sufficient for laser shock peening of Cu–Zn alloys [32]. The morphology of the welding area from solidifying aluminium is also consistent with the assumption of the shock peening process. The grains become ultrafine under the impact of the strong shock wave. The increase in the hardness will also enhance the strength.

The total dynamics may be also a cyclic process of the generation of a small piece of the molten pool with plasma when each pulse comes and the subsequent cooling and solidification of this instantaneous molten pool simultaneously accompanied by the laser peening if the laser intensity is properly controlled. The high strain rates caused by an fs pulse do not always satisfy the fragmentation criteria because of its short duration (tens of picoseconds) [29]. It is possible that the permanent molten pool can exist in a long and very narrow region along/around the axis of the laser beam at the high power. We

conjecture that the molten bilateral glass can be regarded as a coarse indication of the upper limit. Therefore, the proper power may be less than 800 mW at most parameter sets since the SEM photograph illustrates that the melt in the glass is useless for the strength.

However, the filament welding generates the cracks in the glass in the case of welding the soda-lime glass and the graphite film (the welding strength is also strong even though there are cracks in the glass). The collapse position passes the interface at $d = -15 \mu\text{m}$ in Ref. [7] and the difference in this critical value of d may result from the different determination of $d = 0$. Graphite is semi-metal. It has free electrons that can linearly absorb the light. In addition, it also has the covalent bonds (sp^2) and the explosion takes place when enough covalent bonds are broken. Therefore, the intensity for laser peening may be much lower than in the case of welding metal. The effect of the shock wave can penetrate a $20 \mu\text{m}$ graphite film [7]. This shows that the effect of the laser peening can also play a role in the usual welding (the focus is outside the glass). So, the influence of the shock wave likewise depends on the interaction of the laser and matter, not only on the intensity.

7.5. A soft front layer: a transient elastic layer hinders the crack

The heat conductivity of soda-lime glass is about $1.05 \text{ Wm}^{-1} \text{ K}^{-1}$ [39] and that of pure aluminium is $237 \text{ Wm}^{-1} \text{ K}^{-1}$ [40]. The high temperature region is at the interface but mostly on the glass side. It results in a thin layer of glass at the front of the instantaneous modification region can melt. This soft glass layer acts like an elastic body. The elastic strain is reversible, so that a crack-free welding can be realized even though the shock wave repeats every time a pulse impinges. The cool feature (low absorption) of the filamentation is a major reason for keeping this soft front layer thin enough (keeping the instantaneous molten pool thin enough).

7.6. Necessity of a physical contact and a normal force

Many papers discussed the maximum gap for a successful welding [11], [16]. However, we always hold an opposite standpoint. Indeed, the optical contact may be not really necessary as our results show. We deem that a physical contact is still required for obtaining a high welding strength. This is because a large gap means that the density decreases and the porosity increases. The manifold defects in the welding region must form during the solidification process, rather than the melting process. Considering the intrinsic brittleness of the glass, we think that the smallest defects in the welding region are better.

The volume always changes when most material melts and the large change in the volume will generate the cracks. The gap will relieve the stress from the view of the whole molten pool. However, our results show there may be another way. The high density can be obtained by controlling the amount of the melting materials during each interaction cycle. The material can sustain a high stress in a very local regime if the volume only changes a little and this change recovers in a short time. A high limitation of the welding region (a small instantaneous molten pool) in each step of the welding process means a high tolerance for the change in the volume. The effect of the different thermal expansion coefficients can be quite relieved. The process is like that the size of the modification region grows piece by piece. The gap becomes unnecessary and detrimental if there is not a large-sized heat liquid phase in the whole welding process. Depending the welding dynamics above-mentioned, the optical contact is not necessary but the physical contact is also important.

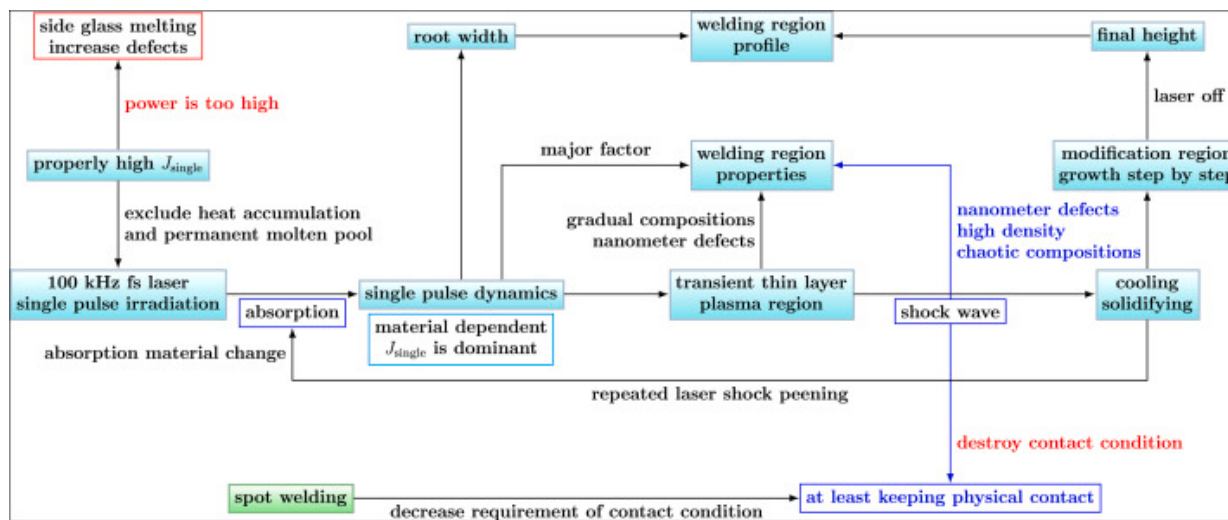
In addition, the shock wave caused by high intensity pulse laser may also open an emergent gap. The spot welding may decrease this impact force because the volume of the weld is much more limited than the scanning welding. This emergent gap can be also hindered by the first weld as the load test shows. A proper design of the laser welding system can uniformly apply a normal force on the sample and can also impede this emergent gap.

7.7. Summary

The high shear strength of our samples should be attributed to the density of the welding region resulting from the heat and cooling as well as soft and hard gradual-step dynamics with repeated laser peening above-discussed. The welding dynamics are very complicated. Nevertheless, the robustness of the welding seems similar whatever the welding mode is. It is partly due to the stability of the nonlinear propagation of the filament.

The chaos of the compositions in the mixed plasma area is impressive. It should arise from the laser peening process under the impact of the shock wave. We do not think that it is a problem since the density is more important than the specific compositions [13].

In summary, we think that there are two major factors that are advantageous to the strength in our experiment. First is the step process due to both a relatively low repetition rate and a relatively high intensity. Second is the effect of the laser peening resulting from the shock wave generated by the fs pulse with a relatively high intensity. Therefore, the physical contact of two materials is at least required and the optical contact should be better. Which one is preferable, filament welding or usual welding, depends on the interactions and the dynamics of the fs laser and the materials. We summarize the major discussions in Fig. 8.



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Fig. 8. The summary of the major discussions. J_{single} is the energy density at the interface.

8. Conclusions

Our experimental results show that a 100 kHz fs laser can perform well in the laser welding even though the heat accumulation is not dominant. The strong strength of an fs laser welding can be obtained if the welding region is formed from the solidifying plasma. Both scanning and spot welding have enough strength if there are only the defects at the nanoscale in the welding region. For avoid the large-sized defects, the physical contact must be achieved even though the optical contact is not necessary. The great challenge of an fs laser welding is still its robustness. The wave shock in the welding process may open an emergent gap and destroy the robustness (break the physical contact condition). The normal force on the welding sample is advantageous for hindering the gap. It is easily implemented if the optical design is upside down (transparent material below and metal above). The spot welding shows a great potential because the welding can be performed at the discrete physical contact areas.

The dynamics of the interaction of a single laser pulse and materials is important for welding when using an fs laser with a relatively low repetition rate. However, the materials which respond to the light change when each pulse impinges. Consequently, the complexity of the welding dynamics in our experiments is much greater than we expected. From practical perspective, our results show that the repeatability of the welding is credible.

CRediT authorship contribution statement

Hongrui Wang: Writing – original draft, Visualization, Investigation. **Wenchao Zhong:** Investigation. **Zizheng Zhao:** Investigation. **Bowen Liu:** Resources. **Dongqing Pang:** Writing – original draft, Supervision, Conceptualization. **Minglie Hu:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Multimedia Component 1.

Data availability

No data was used for the research described in the article.

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